FPGA-Based Fault Detection, Isolation, and Protection (FDIR) System for a 3U CubeSat

Engineering Design, RTL Implementation, and Verification Report

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1 Abstract

Small CubeSats operate with limited power, computing resources, and hardware redundancy, making autonomous fault detection and protection important for reliable operation. This project presents an FPGA-based Fault Detection, Isolation and Protection (FDIR) system for a representative 3U CubeSat. The proposed system continuously monitors important health parameters, including battery voltage, 5 V power rail, payload current, FPGA temperature, OBC heartbeat, communication status, and payload status. Persistent abnormal conditions are detected using threshold checking, persistence mechanisms, and watchdog logic, while a fault manager classifies faults according to their severity.

A protection finite-state machine (FSM) controls the system response by disabling affected functions, entering safe mode for critical conditions, and maintaining critical faults in a latched state until an authorized fault-clear operation is received. An 8-bit fault register is used to record detected fault conditions, and a UART interface is included for fault-status and fault-clear interaction. The complete design was implemented in RTL and verified using behavioral simulation with fault injection. Seven required test cases, including normal operation, persistent over-current, transient current spike, OBC heartbeat failure, over-temperature, multiple faults, and fault clearing, were executed successfully.

2 Introduction

Small satellites such as CubeSats operate with limited power, computing resources, and hardware resources. Because of these limitations, a fault in an important subsystem can affect the overall operation of the spacecraft. An autonomous Fault Detection, Isolation and Protection (FDIR) mechanism can continuously monitor spacecraft health and take predefined protective actions when abnormal conditions persist. The design considers a representative 3U CubeSat containing an On-Board Computer (OBC), Electrical Power System (EPS), communication subsystem, payload, and Attitude Determination and Control System (ADCS).

In this project, an FPGA-based FDIR controller is designed as an independent supervisory system. The FPGA monitors important health parameters such as battery voltage, 5 V power-rail voltage, payload current, FPGA temperature, OBC heartbeat, communication status, and payload status. The objective is to detect abnormal conditions in real time and generate appropriate protection actions without depending completely on the main OBC software.

The design uses threshold-based fault detection together with persistence/debounce logic to prevent short-duration disturbances from being incorrectly treated as faults. An OBC watchdog is used to detect the loss of the OBC heartbeat. Detected faults are recorded in a fault register and classified according to their severity. The protection FSM then performs the required response, such as disabling the affected payload or entering a safe state for critical faults. Critical faults are latched until an authorized fault-clear operation is performed.

The complete system was implemented at the RTL level and verified through behavioral simulation and fault injection. The verification focused on seven required scenarios: normal operation, persistent payload over-current, transient current spike, OBC heartbeat failure, FPGA over-temperature, multiple simultaneous faults, and fault clearing. All seven test cases were successfully passed in the simulation environment. The assignment scope for this implementation is limited to the FPGA/RTL design and simulation.

3 System Architecture

The system architecture of the FPGA-based Fault Detection, Isolation, and Protection (FDIR) system is structured as an independent supervisory controller for a 3U CubeSat. The primary objective of the architecture is to continuously process critical spacecraft health parameters and trigger deterministic protection actions in hardware.

The architecture comprises the following key functional blocks:

- **CubeSat Health Parameters & Sensor Interface**: Collects and conditions real-time digitized telemetry inputs from core subsystems (power rails, payload currents, system temperature, OBC heartbeat, and communication status lines).
- **Fault Detection Logic**: Executes continuous threshold evaluation and persistence debouncing to eliminate transient spikes from triggering false protection events.
- **OBC Watchdog**: Monitors periodic software pulses from the main On-Board Computer to detect system lockups or crashes.
- **Fault Manager**: Evaluates all active fault conditions simultaneously and assigns severity rankings to prioritize emergency responses.
- **Protection FSM**: A finite state machine that drives system recovery states and controls critical hardware output lines including `PAYLOAD_ENABLE` and `SAFE_MODE`.
- **Fault Register & UART Interface**: Latches detected fault events into an 8-bit status register and communicates system state or processes fault-clearing commands via an asynchronous serial link.

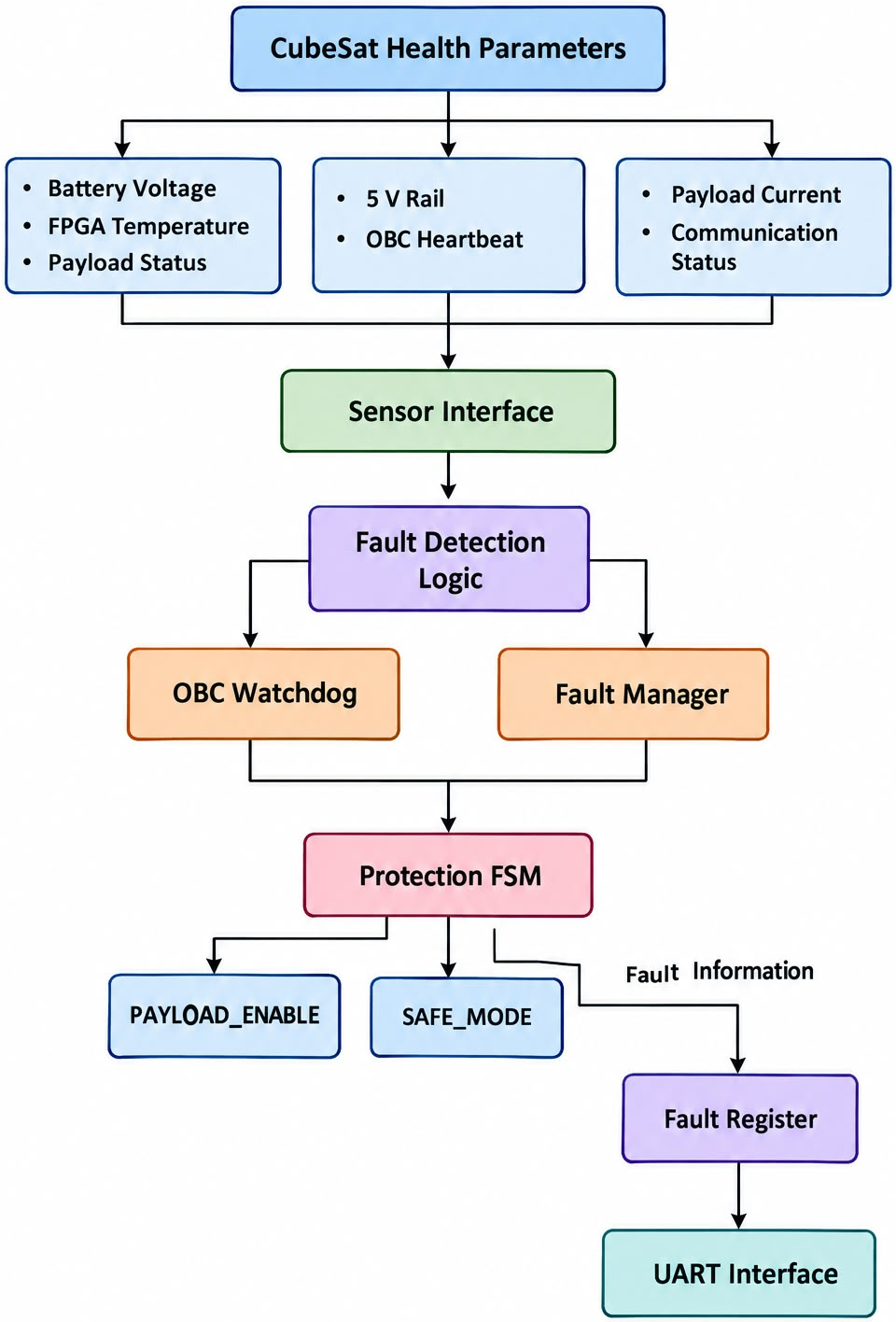


Figure 1: FPGA-Based CubeSat FDIR System Architecture Block Diagram.

4 Monitored Parameters & Fault Conditions

The FDIR system continuously monitors core health parameters across spacecraft subsystems. To ensure reliable fault classification and prevent false triggers from short-duration electrical noise, each parameter is assigned defined normal operating boundaries, fault trip thresholds, and corresponding hardware protection responses.

Table 1: Monitored Health Parameters, Fault Conditions, and System Protection Actions

Parameter	**Normal Condition**	**Fault Condition**	**Protection Action**
Battery Voltage	7.0–8.4 V	$< 7.0\,\text{V}$ or $> 8.4\,\text{V}$	Safe mode / disable non-critical loa
5 V Power Rail	4.75–5.25 V	Outside this range	Disable affected subsystem
Payload Current	$< 2\,\text{A}$	$> 2\,\text{A}$ persistently	Disable payload
FPGA Temperature	$< 85\,^{\circ}\text{C}$	$> 85\,^{\circ}\text{C}$	Thermal protection action
OBC Heartbeat	Valid heartbeat	Heartbeat timeout	Safe-state action
Communication	Valid frames/ACKs	Timeout/invalid comm	Communication fault flagging
Payload Status	Healthy	Failure/unhealthy status	Disable or reset payload

5 Fault Persistence / Debouncing

Fault persistence logic is implemented within the fault detection pipeline to prevent short-duration disturbances, switching noise, or transient electrical spikes from triggering false system trips.

The debouncing process operates according to the following principles:

1. **Continuous Monitoring**: An abnormal telemetry condition (such as an out-of-bounds voltage or current reading) is monitored continuously upon crossing the primary threshold.
2. **Persistence Window Evaluation**: The abnormal condition must persist uninterrupted for a predefined window of clock cycles before the system confirms a valid fault event.
3. **Transient Rejection**: If the signal returns to a nominal operating value prior to reaching the persistence threshold, the internal counter resets and the condition is ignored.

This mechanism prevents unnecessary protective shutdowns and preserves overall mission uptime. For example, if the payload current exceeds 2 A for a duration shorter than the persistence counter threshold, no fault flag is generated. However, if the current remains strictly above 2 A across the entire persistence duration, an over-current fault is confirmed, latched, and forwarded to the fault manager.

6 Fault Severity

To ensure an appropriate and proportional response to spacecraft anomalies, detected faults are classified into three distinct severity levels:

- **Level 1 – Warning**: Indicates an operating condition approaching a safety threshold (e.g., elevated FPGA temperature). A warning status flag is generated and logged, but no immediate power shutdown or load shedding is executed.
- **Level 2 – Recoverable**: Indicates a localized subsystem failure (e.g., payload over-current or communication timeout). The affected subsystem is immediately disabled to prevent hardware damage, with recovery attempted once the condition clears or an authorized command is received.

- **Level 3 – Critical**: Indicates a severe mission-threatening anomaly (e.g., battery undervoltage or OBC watchdog failure). Immediate protection action is enforced: all non-essential loads are cut off, and the spacecraft enters Safe Mode.

6.1 Multi-Fault Prioritization

When multiple fault conditions occur simultaneously, the fault manager evaluates all active signals and automatically prioritizes the highest severity level. Critical (Level 3) protection responses override Level 2 and Level 1 actions to ensure primary spacecraft survivability.

7 RTL Design

The FPGA-based FDIR controller is designed using a modular, hierarchical Verilog RTL structure. Each sub-module performs a dedicated function within the detection, isolation, and recovery pipeline:

- **Sensor Interface**: Receives and conditions incoming health parameters from the spacecraft hardware and analog-to-digital domain.
- **Fault Detector**: Performs threshold checks against predefined upper and lower safety bounds while applying persistence debouncing filters.
- **Watchdog**: Continuously monitors the periodic pulse train from the OBC heartbeat and detects missing edge timeouts.
- **Fault Manager**: Aggregates all active fault inputs from the detector and watchdog modules to evaluate and assign overall system severity.
- **Protection FSM**: Executes the state transitions required to drive control lines (`PAYLOAD_ENABLE`, `SAFE_MODE`) based on the evaluated fault severity.
- **Fault Register**: Stores and latches active fault bits to maintain a persistent record of all system anomalies.
- **UART Interface**: Handles asynchronous serial communication at 9600 baud for external fault-status telemetry transmission and authorization of fault-clear byte commands.
- **Top Level Wrapper**: Structurally interconnects all sub-modules, clock lines, resets, and I/O signals into a unified hardware architecture.

The complete RTL architecture was compiled and verified using a comprehensive testbench environment featuring targeted fault injection scenarios.

8 Fault Register & UART Interface

8.1 8-Bit Fault Register

An 8-bit latched register is used to store and maintain active fault states across spacecraft operations. Each bit represents a specific subsystem fault condition:

Critical faults remain latched in the status register once triggered, ensuring that temporary clearing of abnormal telemetry does not prematurely clear safe mode until an explicit, authorized command is issued.

Table 2: 8-Bit Fault Register Mapping

Bit Position	Fault Condition	Severity Level
Bit 0	Battery Under-Voltage	Level 3 (Critical)
Bit 1	Battery Over-Voltage	Level 3 (Critical)
Bit 2	Payload Over-Current	Level 2 (Recoverable)
Bit 3	FPGA Over-Temperature	Level 1 (Warning)
Bit 4	OBC Watchdog Timeout	Level 3 (Critical)
Bit 5	Communication Timeout	Level 2 (Recoverable)
Bit 6	Payload Failure	Level 2 (Recoverable)
Bit 7	Reserved	N/A

8.2 UART Interface

The asynchronous UART interface operates at 9600 baud and provides a dedicated communication channel between the FDIR FPGA controller and external ground stations or host controllers. It performs the following primary functions:

- Transmitting real-time 8-bit fault register status frames.
- Receiving external command bytes for system control.
- Decoding the authorized `FAULT_CLEAR` byte (`0xCC`) to reset latched fault flags after an anomaly has been assessed and isolated.

9 Protection FSM

The Protection Finite State Machine (FSM) serves as the core decision-making unit of the FDIR controller. It evaluates confirmed fault conditions from the fault manager and drives appropriate hardware control outputs to safeguard spacecraft operations.

9.1 FSM State Transitions

The Protection FSM transitions sequentially across six distinct operational states:

1. **INIT**: Sets default initial conditions, clears temporary counters, resets internal state registers, and asserts nominal enable signals (`payload_enable` = 1).
2. **MONITOR**: Continuously samples telemetry inputs and monitors the system for out-of-bounds parameter conditions or watchdog timeouts.
3. **FAULT ANALYSIS**: Evaluates confirmed persistent anomalies and classifies the active fault condition into Level 1 (Warning), Level 2 (Recoverable), or Level 3 (Critical).
4. **PROTECT**: Executes targeted protection responses for Level 2 faults by disabling specific non-critical functions or cutting power to the affected subsystem.
5. **SAFE MODE**: Enforces emergency spacecraft protection upon detecting Level 3 critical faults by opening load switches, isolating payloads, and maintaining essential bus power.
6. **RECOVERY / LATCH**: Evaluates fault type post-isolation; recoverable faults return to `MONITOR` state once cleared, whereas critical faults transition to a locked `LATCH` state until an authorized `FAULT_CLEAR` command byte (`0xCC`) is received via UART.

10 Verification & Testbench

The RTL design was verified using a self-checking testbench in Xilinx Vivado. The verification environment includes a automated stimulus generator that simulates real-time spacecraft sensor telemetry, OBC heartbeat pulses, and external UART command transmissions.

To thoroughly validate the system, targeted fault injection was used to simulate normal operation, transient noise, persistent over-current events, temperature spikes, watchdog lockups, multiple concurrent failures, and recovery commands.

11 Test Cases & Results

A total of seven specific test scenarios were executed to validate the FDIR controller logic. All test cases completed successfully during RTL behavioral simulation.

Table 3: RTL Behavioral Simulation Test Cases and Results

Test #	Test Case Scenario	Observed System Response	Status
1	Normal Operation	No fault detected; `PAYLOAD_ENABLE` remains active	PASS
2	Persistent Payload Over-Current	Over-current confirmed; payload disabled	PASS
3	Transient Current Spike	Short-duration spike filtered out; payload remains active	PASS
4	OBC Heartbeat Failure	Watchdog timeout detected; critical fault flagged	PASS
5	FPGA Over-Temperature	Thermal anomaly detected; Level 1 fault flagged	PASS
6	Multiple Faults	Highest severity level selected; safe mode enforced	PASS
7	Fault Clear Command	Latched fault status register cleared via UART	PASS

12 Simulation Waveforms

Behavioral simulation waveforms were analyzed in Xilinx Vivado to verify timing behavior, state transitions, and signal integrity under various operational conditions. The waveforms capture real-time transitions across key input parameters and internal controller outputs, including:

- **Input Telemetry Signals**: Battery voltage, 5 V power rail voltage, payload current, and FPGA temperature.
- **Status & Control Signals**: OBC heartbeat, communication status lines, and payload operational status.
- **Internal Flag Registers**: Bitwise 8-bit fault status register state.
- **Hardware Output Lines**: `PAYLOAD_ENABLE`, `SAFE_MODE`, and temperature warning indicators.

The simulation waveform confirms that the FDIR system detects the injected faults, latches confirmed fault flags, and generates the corresponding hardware protection response within the deterministic time window following fault confirmation.

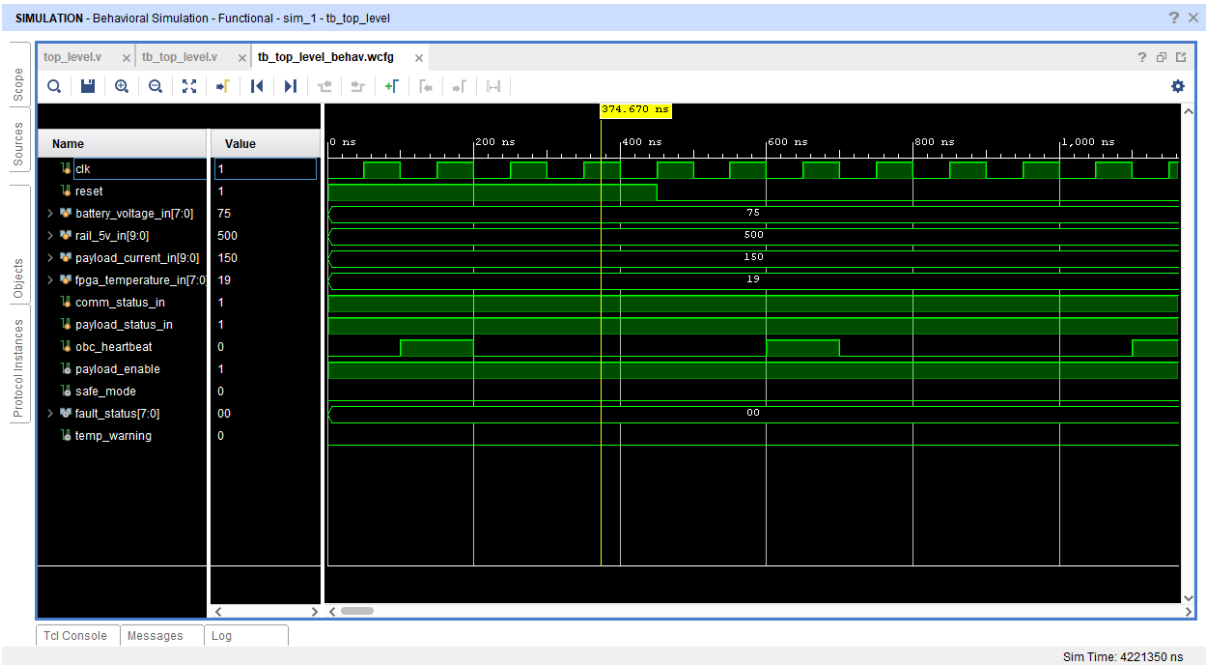


Figure 2: Simulation waveform showing monitored parameters, fault status, and protection outputs.

13 Results & Discussion

The FPGA-based FDIR controller was verified through comprehensive RTL behavioral simulation using seven distinct test scenarios. The simulation results confirm that the system reliably detects anomalous conditions, applies appropriate debouncing filters, and asserts the necessary hardware protection outputs.

- **Normal Operation**: Evaluated nominal telemetry inputs without asserting false positive fault flags; `PAYLOAD_ENABLE` remained continuously active.
- **Persistent Payload Over-Current**: Confirmed that sustained current exceeding 2 A across the persistence window successfully triggers payload isolation.
- **Transient Current Spike**: Demonstrated that high-amplitude current spikes lasting less than the persistence threshold are correctly filtered out without interrupting subsystem operation.
- **OBC Heartbeat Failure**: Verified that a missing heartbeat pulse train triggers the watchdog timer, asserting a Level 3 critical fault flag.
- **FPGA Over-Temperature**: Validated thermal threshold monitoring by flagging an over-temperature condition when simulated temperature exceeded 85 °C.
- **Multiple Concurrent Faults**: Confirmed multi-fault prioritization by verifying that the fault manager automatically asserts Level 3 Safe Mode when simultaneous faults occur.
- **Fault Clear Operation**: Demonstrated successful decoding of the authorized UART clear byte (`0xCC`), resetting latched fault register bits post-isolation.

Overall, the simulation results demonstrate the robust, deterministic operation of the proposed FPGA-based FDIR architecture across all defined spacecraft fault conditions.

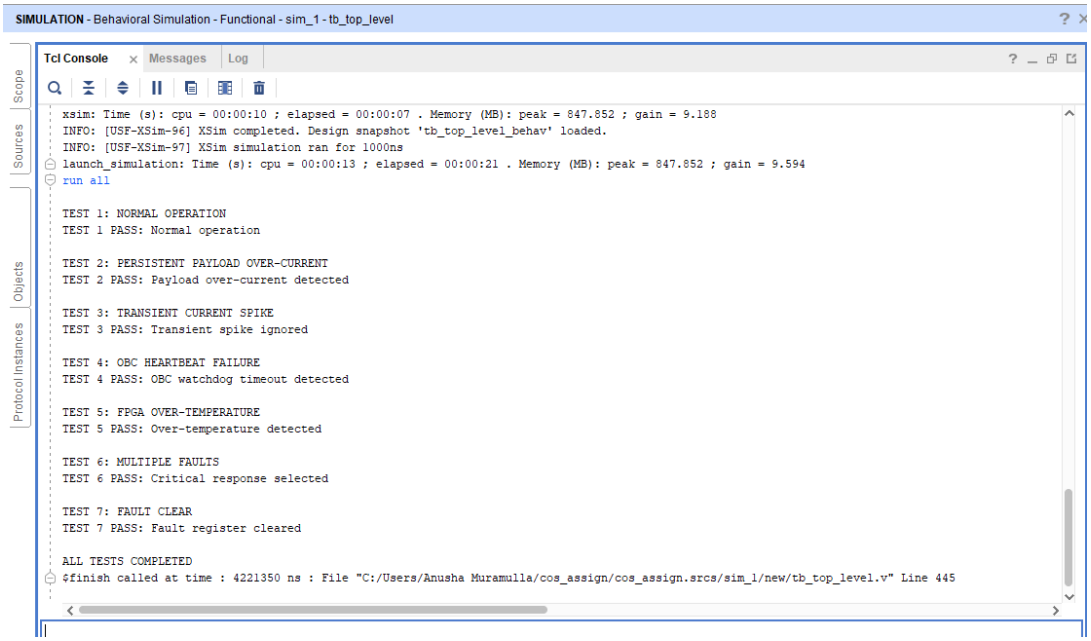


Figure 3: Testbench output showing successful completion of all seven test cases.

14 Assumptions & Limitations

14.1 Design Assumptions

The implementation and verification of the FPGA-based FDIR controller rely on the following design assumptions:

- **Operational Thresholds**: Monitored parameter bounds and trip thresholds are established based on representative 3U CubeSat health parameters.
- **Digitized Telemetry Inputs**: Sensor values and health metrics are supplied to the core logic as pre-conditioned digital inputs during RTL simulation.
- **Persistence Debouncing**: Fault conditions are confirmed and passed to the fault manager only after remaining active continuously across the predefined persistence window.
- **Critical Fault Latching**: Critical (Level 3) fault states remain latched in the status register until an explicit, authorized `FAULT_CLEAR` command byte is received.

14.2 System Limitations

The current scope of the project presents the following limitations:

- **RTL-Only Verification**: System operation was verified strictly through RTL behavioral simulation in Xilinx Vivado; physical FPGA board deployment and hardware-in-the-loop (HIL) testing were not conducted.
- **Abstracted Physical Interfaces**: Analog front-end circuits, real-world sensors, and physical Analog-to-Digital Converter (ADC) interfaces were not included in the RTL scope.
- **Simplified Communication Protocol**: Serial communication is modeled at the functional RTL register level rather than integrating a full space-grade telemetry and telecommand stack.

- **Scope Boundary**: The controller serves as a representative FDIR architecture and does not encompass full flight-qualified hardware or mission-specific environmental testing.

15 Conclusion

An autonomous FPGA-based Fault Detection, Isolation, and Protection (FDIR) system was successfully designed and verified at the RTL level for a representative 3U CubeSat architecture. The developed system continuously monitors vital health parameters, applies persistence filtering to reject transient electrical noise, classifies anomalies according to a three-tier severity scheme, latches detected fault states into an 8-bit status register, and drives deterministic hardware protection signals.

The system's functional correctness was validated through behavioral simulation in Xilinx Vivado across seven comprehensive test cases: normal operation, persistent payload overcurrent, transient current spikes, OBC heartbeat watchdog timeout, FPGA over-temperature, multiple simultaneous fault prioritization, and command-driven fault clearing. All test cases passed successfully, confirming that the FPGA-based supervisory architecture operates reliably without depending entirely on software-level OBC intervention.